24 - Model Selection; Greedy Methods, Stepwise Selection. Shrinkage Methods: PCA Regression, Ridge Regression, Lasso. Regression with Penalizations. Generalized Least Squares (GLS).

Applied Statistics

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1 Big Picture — Why Do We Need These Methods?

- In linear regression we have a response y and r potential predictors $z_1, z_2, \dots, z_r$.
- **Problem 1:** Too many variables — which ones should enter the model?
- **Problem 2:** Collinearity — predictors are highly correlated with each other, making OLS estimates unstable and exploding in variance.
- **Two families of solutions:**
  - **Model/variable selection** — decide which variables to include or exclude.
  - **Shrinkage / regularisation** — keep all variables but penalise large coefficients (PCA regression, Ridge, Lasso).
- The design matrix is $\mathbb{Z} = [\mathbf{1}\ \underset{\sim}{z}_1\ \cdots\ \underset{\sim}{z}_r]$ with $\underset{\sim}{z}_j \in \mathbb{R}^n$.
- With r predictors there are 2^r possible linear models — too many to enumerate for large r.

2 Model (Variable) Selection

2.1 Best Subset (Greedy) Selection

- **Goal:** Find the single best model for each size $k = 0, 1, \dots, r$.
- **Algorithm:** For each k, fit all $\binom{r}{k}$ models with exactly k regressors, compute R^2, keep the best. Call it R^2_k.
- The sequence satisfies:

$$R^2_0 \le R^2_1 \le \cdots \le R^2_r$$

- **How to choose k:** Plot R^2_k, $R^2_{\text{adj},k}$, AIC, or BIC against k. Find the **elbow** — the point where adding another variable stops giving meaningful improvement.
- **Note:** To use AIC, data must be Gaussian (AIC is based on Gaussian likelihood by default).
- **Limitation:** Total models to evaluate $= \sum_{k=0}^{r} \binom{r}{k} = 2^r$. For $r = 40$ this is over 10^{12} — computationally impossible.

2.2 Stepwise Iterative Selection

- **Goal:** Find a *good* model (not necessarily optimal) efficiently.
- Two directions: **forward** (add one at a time) or **backward** (remove one at a time).

2.2.1 Forward Selection

- Start with the best 1-regressor model. Repeat until convergence:
  - Add the variable that most improves the fit.
  - Test whether the improvement is statistically significant using the **F-test**.
- **F-test for nested models** — suppose $k' > k$:

$$F = \frac{\dfrac{SS_{\text{res}}(\mathbb{Z}_k) - SS_{\text{res}}(\mathbb{Z}_{k'})}{k' - k}}{\dfrac{SS_{\text{res}}(\mathbb{Z}_{k'})}{n - (k' + 1)}}$$

  - Numerator: how much residual variance the new variables explain, normalised by the number added.
  - Denominator: residual variance of the larger model.
  - **Stop** when the p-value is large — adding more variables does not pay off.
- **Advantage:** Forward selection avoids the collinearity problem because variables are added one at a time; a second correlated variable adds little after the first is already in.

2.2.2 Backward Selection

- Start with the full model (all r regressors). At each step, remove the variable that least hurts the fit.
- Stop when removal starts hurting significantly (same F-test logic).
- **Variants:** Add 2 variables and remove 1; add 3 and remove 2; etc.
- **Important constraint:** The iterative approach works only if models are **nested** (one model's predictor set is a subset of the other's).
- **Modern alternatives:** PCA regression, Ridge, Lasso — these tell you *why* a variable should be removed.

3 Centring the Data

- OLS model: $\underline{y} = \mathbb{Z}\underline{\beta} + \underline{\epsilon}$, with solution $\underline{\hat{\beta}} = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y}$ (assuming $\mathbb{Z}$ has full rank).
- The fitted model always passes through the **barycentre** of the data:

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1\bar{z}_1 + \cdots + \hat{\beta}_r\bar{z}_r \implies \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{z}_1 - \cdots - \hat{\beta}_r\bar{z}_r$$

- The fitted model can be rewritten as:

$$y_0 - \bar{y} = \hat{\beta}_1(z_1 - \bar{z}_1) + \cdots + \hat{\beta}_r(z_r - \bar{z}_r)$$

- **Centring:** Define $\underline{y}^* = [y_1 - \bar{y}, \ldots, y_n - \bar{y}]^T$ and the centred design matrix:

$$\mathbb{Z}^* = \begin{bmatrix} z_{11} - \bar{z}_1 & \cdots & z_{1r} - \bar{z}_r \\ \vdots & & \vdots \\ z_{n1} - \bar{z}_1 & \cdots & z_{nr} - \bar{z}_r \end{bmatrix} \in \mathbb{R}^{n \times r}$$

- The OLS problem on centred data (no intercept column needed):

$$\underline{\hat{\beta}}^* = \underset{\underline{\beta} \in \mathbb{R}^r}{\arg\min} \|\underline{y}^* - \mathbb{Z}^*\underline{\beta}\|^2$$

- After fitting, recover $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{z}_1 - \cdots - \hat{\beta}_r\bar{z}_r$.
- **From now on** we assume data is centred and drop the $*$ superscript.

4 Collinearity and Variable Selection: Three Modern Methods

4.1 PCA Regression

- **Problem:** Collinearity means columns of $\mathbb{Z}$ are nearly linearly dependent $\Rightarrow \mathbb{Z}^T\mathbb{Z}$ is nearly singular $\Rightarrow$ OLS estimates explode.
- **Idea:** Transform the correlated regressors into uncorrelated ones via PCA, then regress on those.
- Apply PCA to $\mathbb{Z} = [\underline{z}_1, \ldots, \underline{z}_r]$ (centred) to obtain principal components:

$$PC_k = e_{1k}z_1 + e_{2k}z_2 + \cdots + e_{rk}z_r, \quad k = 1, \ldots, r$$

where the e_{ij} are eigenvectors (loadings) of $\mathbb{Z}^T\mathbb{Z}$.

- **Key property:** $PC_1, \ldots, PC_r$ are orthogonal (uncorrelated) — collinearity is solved.
- **Dimension reduction:** Keep only the first $k \leq r$ components (chosen via scree plot elbow). New design matrix: $\mathbb{Z}^* = [PC_1, \ldots, PC_k] \in \mathbb{R}^{n \times k}$.
- Fit the reduced model:

$$\underline{y} = \mathbb{Z}^*\underline{\beta} + \underline{\epsilon}, \quad \underline{\beta} \in \mathbb{R}^k$$

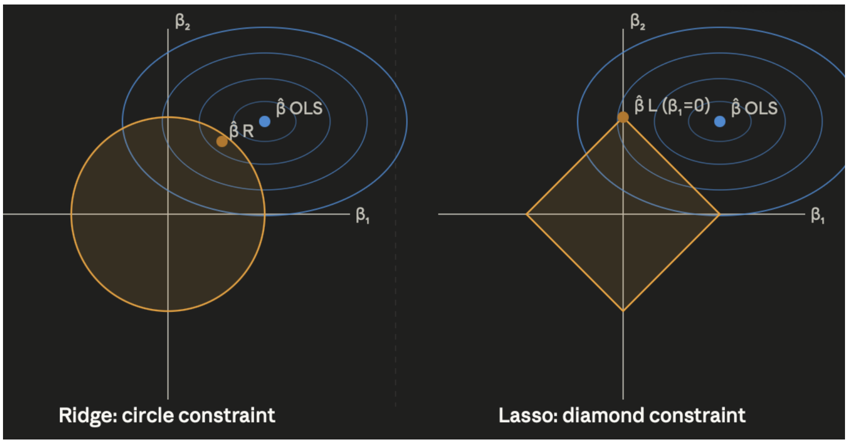

Figure 1: Enter Caption

- Substituting back:
$$y_0 = z_1\hat{\gamma}_1 + \cdots + z_r\hat{\gamma}_r$$
where $\hat{\gamma}_i = \sum_{j=1}^{k} e_{ij}\hat{\beta}_j$ — all original variables remain with nonzero coefficients.
- **Limitation 1:** PCA maximises variance in $\mathbb{Z}$, not predictive power for y — you may discard components that are informative for y.
- **Limitation 2:** The solution is *not sparse* — no variable selection.
- **Note:** Algorithms that do dimension reduction while preserving correlation with y exist (e.g. slice-inverse regression).

4.2 Ridge Regression (Hoerl & Kennard)

- **Problem:** Collinearity makes $\mathrm{Var}(\hat{\beta}_i)$ explode. We want to keep coefficients small and stable.
- **Constrained formulation:**
$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 \quad \text{subject to} \quad \|\underline{\beta}\|^2 \le s$$
- **Equivalent Lagrangian (penalised) formulation:**
$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 + \lambda\|\underline{\beta}\|^2$$
where $\lambda \ge 0$ is the Lagrange multiplier (penalty parameter), a function of s.
- **Closed-form solution:**
$$\hat{\underline{\beta}}_R = (\mathbb{Z}^T\mathbb{Z} + \lambda I)^{-1}\mathbb{Z}^T\underline{y}$$
Compare to OLS: $\hat{\underline{\beta}}_{\text{OLS}} = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y}$. Adding λI to the diagonal makes the matrix always invertible — this directly fixes the collinearity problem.
- **Geometric intuition:** OLS contours are ellipses centred at $\hat{\underline{\beta}}_{\text{OLS}}$. The constraint $\|\underline{\beta}\|^2 \le s$ is a sphere. The Ridge solution $\hat{\underline{\beta}}_R$ is where the smallest ellipse is tangent to the sphere.

- As $s \to 0$: $\lambda \to \infty$, solution shrinks to $\mathbf{0}$ (no regression). As $\lambda = 0$: OLS solution.
- **Properties:**
  - $\underline{\hat{\beta}}_R$ is a **biased** estimator of $\underline{\beta}$.
  - But there always exists $\lambda > 0$ such that the MSE is lower:

$$\mathbb{E}\left[\|\underline{\hat{\beta}}_R - \underline{\beta}\|^2\right] < \mathbb{E}\left[\|\underline{\hat{\beta}}_{\text{OLS}} - \underline{\beta}\|^2\right]$$

  - Higher bias, but lower variance — the **bias-variance trade-off**.
  - $\underline{\hat{\beta}}_R$ is called a **shrinkage estimator**: it shrinks OLS toward zero.
  - **Does NOT perform variable selection** — no coefficient is set exactly to zero.
- **Choosing λ:** Cross-validation — find the λ that minimises prediction error for y.
- **Important:** $\underline{\hat{\beta}}_R$ is *not* scale-invariant. You must **standardise** all regressors and the response before fitting Ridge.
- When regressors are orthogonal after standardisation, $\underline{\hat{\beta}}_R$ is simply $\underline{\hat{\beta}}_{\text{OLS}}$ re-scaled (shrunk by $1/(1+\lambda)$).

4.3 Lasso Regression (Tibshirani)

- **Idea:** Same as Ridge but replace the L^2 norm constraint with an L^1 norm constraint:

$$\arg\min_{\underline{\beta}} (\underline{\beta} - \underline{\hat{\beta}})^T \mathbb{Z}^T \mathbb{Z} (\underline{\beta} - \underline{\hat{\beta}}) \quad \text{subject to} \quad \|\underline{\beta}\|_1 = \sum_{i=1}^{r} |\beta_i| \leq s$$

- **Lagrangian formulation:**

$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 + \lambda \|\underline{\beta}\|_1$$

- **Geometric intuition (key difference from Ridge):** The L^1 ball is a **diamond** (rotated square in 2D). Its corners lie exactly on the axes. The OLS ellipses are most likely to first touch the diamond at a corner, where one or more $\beta_i = 0$ exactly. This is feature selection by geometry.
- **Properties:**
  - Biased estimator — higher bias, lower variance (bias-variance trade-off).
  - **No closed-form solution** — must be solved numerically.
  - **Performs automatic variable selection**: some $\hat{\beta}_i = 0$ exactly $\Rightarrow$ those variables are removed.
  - Handles collinearity and model selection simultaneously.
- **Choosing λ:** Cross-validation on prediction error for y.

4.4 Generalisations: L^q Norms and Elastic Net

- You can generalise the constraint to any L^q norm:

$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 \quad \text{with} \quad \|\underline{\beta}\|_q = \left(\sum_i |\beta_i|^q\right)^{1/q} \leq s$$

  - $q = 2$: Ridge (sphere constraint, smooth, no corners).
  - $q = 1$: Lasso (diamond constraint, corners on axes, variable selection).
  - $q < 1$: Even spikier — stronger variable selection, but problem is **non-convex** and hard to optimise.
  - $q \to 0$: Equivalent to pure model selection (counting non-zero coefficients).
- **Elastic Net** — combine Ridge and Lasso simultaneously:

$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\| + \lambda_1\|\underline{\beta}\|_1 + \lambda_2\|\underline{\beta}\|_2^2$$

  - Pure machine-learning territory: no uncertainty measures, no distributional properties — just prediction.
  - Works well when background knowledge suggests a particular type of solution.
  - Overall idea: fit a regression plus a penalisation.

5 Connections with GLM (Generalised Linear Models)

- All methods so far assumed y is continuous and Gaussian. What if y is binary (0/1), a count, or strictly positive?
- The basic model is $y = f(\underline{z}) + \epsilon$.
  - Linear regression: $f(\underline{z}) = \mathbb{E}[y|\underline{z}] = \beta_0 + \beta_1 z_1 + \cdots + \beta_r z_r$, and $\mathbb{E}[\epsilon] = 0$, $\mathrm{Var}[\epsilon] = \sigma^2$.
  - Gaussian assumption: $\epsilon \sim \mathcal{N}(0, \sigma^2) \Rightarrow y|\underline{z} \sim \mathcal{N}(f(\underline{z}), \sigma^2)$.
- **GLM generalisation:** Apply a **link function** g to $\mathbb{E}[y|\underline{z}]$:

$$g(\mathbb{E}[y|\underline{z}]) = \beta_0 + \beta_1 z_1 + \cdots + \beta_r z_r$$

  and assume $y|\underline{z} \sim F$ for some distribution F (Poisson, Bernoulli, Gaussian, ...).
- The link function maps a constrained quantity (e.g. probability in $(0, 1)$) to the real line where the linear predictor lives.
- **Common link functions:**
  - $Y \sim \mathcal{N}$: $g(\mu) = \mu$ (identity) $\Rightarrow$ *Linear Regression.*
  - $Y \sim \mathrm{Be}(p)$: $g(p) = \mathrm{logit}(p) = \log\left(\frac{p}{1-p}\right) \Rightarrow$ *Logit (Logistic) Regression.*
  - $Y \sim \mathrm{Poisson}(\lambda)$: $g(\lambda) = \log(\lambda) \Rightarrow$ *Poisson Regression.*

  - $Y \sim \text{Be}(p)$: $g(p) = \Phi^{-1}(p)$, where Φ is the CDF of $\mathcal{N}(0,1) \Rightarrow$ *Probit Regression.*

- **Cost:** Least squares no longer works. Parameters $\underline{\beta}$ are estimated by **maximum likelihood (MLE)**, solved numerically.
- **What we lose:** No closed-form estimator; no simple variance formulas; no distributional properties for the estimators.
- **What carries over:** Ridge and Lasso penalties can be added directly to the negative log-likelihood — all shrinkage and variable selection intuitions apply in GLM.
- **Note:** Since we use maximum likelihood (not least squares), we don't have variance properties nor closed-form properties for the estimators.

Method Summary Table

Method	Solves collinearity	Variable selection	Closed form	Biased
Best subset	Partial	Yes (optimal)	Yes	No
Stepwise	Partial	Yes (good)	Yes	No
PCA regression	Yes	No	Yes	No (if all PCs kept)
Ridge	Yes	No	Yes	Yes
Lasso	Yes	Yes	No	Yes

Türkçe Özet

Aynı konuların Türkçe açıklaması

6 Büyük Resim — Neden Bu Yöntemlere İhtiyaç Var?

- Lineer regresyonda y bir yanıt değişkeni, $z_1, \ldots, z_r$ ise aday tahmin edici değişkenlerdir.
- **Problem 1:** Çok fazla değişken var — hangilerini modele alalım?
- **Problem 2:** Collinearity (çoklu doğrusallık) — tahmin ediciler birbirine çok benziyorsa OLS katsayıları dengesiz ve çok büyük çıkar.
- **İki çözüm ailesi:**
  - **Model/değişken seçimi** — hangi değişkenlerin içeride, hangilerinin dışarıda olduğuna karar ver.
  - **Büzme / düzenlileştirme** — tüm değişkenleri tut ama büyük katsayıları cezalandır (PCA regresyonu, Ridge, Lasso).
- r değişkenle 2^r farklı model mümkündür — büyük r için tümünü denemek imkânsızdır.

7 Model (Değişken) Seçimi

7.1 En İyi Alt Küme (Greedy) Seçimi

- **Hedef:** Her $k = 0, 1, \ldots, r$ boyutu için en iyi modeli bul.
- Her k için tam olarak k değişken kullanan $\binom{r}{k}$ modeli kur, R^2 hesapla, en iyisini sakla (R_k^2).
$$R_0^2 \leq R_1^2 \leq \cdots \leq R_r^2$$
- R_k^2, $R^2_{\text{adj},k}$, AIC veya BIC grafiğini çiz; **dirsek (elbow)** noktasını bul.
- **Sınırlama:** Toplam 2^r model — r büyükse hesaplanamaz.

7.2 Adım Adım (Stepwise) Seçim

- **Hedef:** Optimal değil, iyi bir model bul; verimli şekilde.
- **İleri (Forward):** Hiçbir değişken yokken başla. Her adımda en fazla iyileştiren değişkeni ekle.
- **Geri (Backward):** Tüm değişkenlerle başla. Her adımda en az katkı yapanı çıkar.
- **Durma kriteri — F-testi** ($k' > k$ için):
$$F = \frac{\dfrac{SS_{\text{res}}(\mathbb{Z}_k) - SS_{\text{res}}(\mathbb{Z}_{k'})}{k' - k}}{\dfrac{SS_{\text{res}}(\mathbb{Z}_{k'})}{n - (k' + 1)}}$$

p-değeri büyükse dur — daha fazla değişken eklemeye değmez.

- **Önemli:** Bu yaklaşım yalnızca **iç içe geçmiş (nested)** modellerde çalışır.
- İleri seçimde collinearity sorunu daha az önemlidir — ikinci korelasyonlu değişken, birincisi zaten içerideyken fazla katkı sağlamaz.

8 Veriyi Merkezleme

- Fitted model her zaman verinin **barycentre**'inden (ortalama noktasından) geçer:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{z}_1 - \cdots - \hat{\beta}_r\bar{z}_r$$

- Merkezlenmiş değişkenlerle ($z - \bar{z}$ ve $y - \bar{y}$) çalışarak kesen terimi ($\hat{\beta}_0$) problemi dışına alabiliriz:
$$\underline{\hat{\beta}}^* = \arg\min_{\underline{\beta}\in\mathbb{R}^r} \|\underline{y}^* - \mathbb{Z}^*\underline{\beta}\|^2$$

- Bundan sonra verilerin merkezlenmiş olduğu varsayılır ve $*$ üst indisi düşürülür.

9 Collinearity ve Değişken Seçimi: Üç Modern Yöntem

9.1 PCA Regresyonu

- **Problem:** Collinearity varsa OLS katsayıları patlıyor.
- **Fikir:** Korelasyonlu değişkenleri birbirinden bağımsız yeni eksenlere (temel bileşenlere) dönüştür, sonra bu yeni eksenler üzerinde regresyon kur.
- PCA uygulanınca elde edilen temel bileşenler:

$$PC_k = e_{1k}z_1 + e_{2k}z_2 + \cdots + e_{rk}z_r$$

- $PC_1, \ldots, PC_r$ birbirinden tamamen **ortogonal (bağımsız)** — collinearity çözülür.
- İlk $k \leq r$ bileşeni tut (scree plot dirseği), yeni tasarım matrisi $\mathbb{Z}^* = [PC_1, \ldots, PC_k]$ ile regresyon kur.
- **Sınırlama 1:** PCA, y'yi değil $\mathbb{Z}$'deki varyansı maksimize eder — y ile ilişkili bilgiyi atıyor olabilirsin.
- **Sınırlama 2:** Çözüm sparse değil — değişken seçimi yapılmaz, tüm z_i'ler son modelde kalır.

9.2 Ridge Regresyon

- **Fikir:** Katsayıların çok büyümesini engelle — modele L^2 normu cezası ekle.
- **Kısıtlı formülasyon:**

$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 \quad \text{s.t.} \quad \|\underline{\beta}\|^2 \leq s$$

- **Lagrangian formu:**
$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 + \lambda\|\underline{\beta}\|^2$$
- **Kapalı form çözüm:**
$$\hat{\underline{\beta}}_R = (\mathbb{Z}^T\mathbb{Z} + \lambda I)^{-1}\mathbb{Z}^T\underline{y}$$
$\mathbb{Z}^T\mathbb{Z}$'nin köşegenine λ eklemek matrisi her zaman terslenebilir yapar — collinearity doğrudan çözülür.
- **Geometrik sezgi:** OLS hata konturları $\hat{\underline{\beta}}_{\text{OLS}}$ merkezli elipsler. Kısıt $\|\underline{\beta}\|^2 \leq s$ bir küredir. Ridge çözümü $\hat{\underline{\beta}}_R$, en küçük elipsin küreye teğet olduğu noktadır.
- **Özellikler:**
  - **Yanlı (biased)** tahmin edici, fakat MSE her zaman daha düşük olabilir:
  $$\mathbb{E}\left[\|\hat{\underline{\beta}}_R - \underline{\beta}\|^2\right] < \mathbb{E}\left[\|\hat{\underline{\beta}}_{\text{OLS}} - \underline{\beta}\|^2\right]$$
  - Yüksek yanlılık, düşük varyans — **yanlılık-varyans dengesi**.
  - **Değişken seçimi yapmaz** — hiçbir katsayı tam sıfıra eşitlenmez.
  - $\hat{\underline{\beta}}_R$ bir **büzme tahmin edicisi** (shrinkage estimator) olarak adlandırılır.
- **λ seçimi:** Çapraz doğrulama (cross-validation).
- **Önemli:** Ridge ölçek değişmezli değildir — kullanmadan önce tüm değişkenler ve yanıt **standartlaştırılmalıdır**.

9.3 Lasso Regresyon

- **Fikir:** Ridge gibi, fakat kısıt L^2 yerine L^1 normu kullanır.
- **Lagrangian formu:**
$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 + \lambda\|\underline{\beta}\|_1, \quad \|\underline{\beta}\|_1 = \sum_{i=1}^{r} |\beta_i|$$
- **Geometrik sezgi (Ridge'den temel fark):** L^1 topu bir **elmas** (2 boyutta köşeli kare). OLS elipsleri büyüdükçe en olası temas noktası köşedir — ve köşede bir ya da daha fazla $\beta_i = 0$ *tam olarak*. Bu geometri üzerinden **değişken seçimidir**.
- **Özellikler:**
  - Yanlı tahmin edici, yüksek yanlılık, düşük varyans.
  - **Kapalı form çözüm yok** — sayısal yöntemle çözülür.
  - **Otomatik değişken seçimi:** Bazı $\hat{\beta}_i = 0$ tam olarak $\Rightarrow$ o değişken modelden çıkar.
  - Collinearity ve model seçimini *aynı anda* çözer.
- **λ seçimi:** y için çapraz doğrulama.

9.4 Genellemeler: L^q Normları ve Elastic Net

- Genel L^q normu kısıtı:

$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 \quad \text{s.t.} \quad \|\underline{\beta}\|_q = \left(\sum_i |\beta_i|^q\right)^{1/q} \leq s$$

  - $q = 2$: Ridge. $q = 1$: Lasso. $q < 1$: Daha güçlü değişken seçimi ama konveks değil. $q \to 0$: Saf model seçimi.

- **Elastic Net** — Lasso ve Ridge'i aynı anda kullan:

$$\arg\min_{\underline{\beta}} \|\underline{y} - \mathbb{Z}\underline{\beta}\| + \lambda_1\|\underline{\beta}\|_1 + \lambda_2\|\underline{\beta}\|_2^2$$

  Makine öğrenmesi dünyasına geçiş: belirsizlik ölçüsü yok, sadece tahmin. Beklenen çözüm tipi hakkında ön bilgi varsa iyi çalışır.

10 GLM ile Bağlantı (Genelleştirilmiş Doğrusal Modeller)

- Şimdiye kadar y'nin sürekli ve Gaussian olduğunu varsaydık. Ya y ikili (0/1), sayım verisi veya kesinlikle pozitifse?
- **GLM genellemesi:** $\mathbb{E}[y|\underline{z}]$'ye bir **bağlantı fonksiyonu** g uygula:

$$g(\mathbb{E}[y|\underline{z}]) = \beta_0 + \beta_1 z_1 + \cdots + \beta_r z_r$$

  ve $y|\underline{z} \sim F$ varsay (Poisson, Bernoulli, Gaussian, ...).

- **Yaygın bağlantı fonksiyonları:**
  - $Y \sim \mathcal{N}$: $g(\mu) = \mu \Rightarrow$ *Lineer Regresyon.*
  - $Y \sim \text{Be}(p)$: $g(p) = \log\left(\frac{p}{1-p}\right) \Rightarrow$ *Lojistik Regresyon.*
  - $Y \sim \text{Poisson}(\lambda)$: $g(\lambda) = \log(\lambda) \Rightarrow$ *Poisson Regresyon.*
  - $Y \sim \text{Be}(p)$: $g(p) = \Phi^{-1}(p) \Rightarrow$ *Probit Regresyon.*
- **Ödediğimiz bedel:** En küçük kareler artık çalışmaz. Parametreler **maksimum olabilirlik (MLE)** ile, sayısal olarak tahmin edilir.
- **Kaybettiklerimiz:** Kapalı form tahmin edici yok; basit varyans formülleri yok; tahmin edicilerin dağılım özellikleri yok.
- **Geçen her şey:** Ridge ve Lasso cezaları negatif log-olabilirliğe doğrudan eklenebilir — tüm büzme ve değişken seçimi sezgileri GLM'de de geçerlidir.

Yöntem Özet Tablosu

Yöntem	Collinearity çözer	Değişken seçimi	Kapalı form	Yanlı
En iyi alt küme	Kısmen	Evet (optimal)	Evet	Hayır
Adım adım seçim	Kısmen	Evet (iyi)	Evet	Hayır
PCA regresyonu	Evet	Hayır	Evet	Hayır (tüm PC'ler tut
Ridge	Evet	Hayır	Evet	Evet
Lasso	Evet	Evet	Hayır	Evet